

The Hyperbolic Structure of Bounded Dynamical Systems: From Evolutionary Trajectories to Measurement Statistics

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Abstract

We show that a single geometric law governs a broad class of *bounded, compositional* one-dimensional dynamics. Under mild axioms (bounded state space with fixed endpoints, C^1 monotone flow, associative time-homogeneous composition, and boundary-fixing homogeneity), there exists a unique increasing *rapidity* coordinate that linearizes the dynamics. By an Aczél-type representation together with a boundary-fixing Möbius action, this linearizer is (up to positive scale) the inverse hyperbolic tangent. Consequently, if $w(t) \in (0, K)$ evolves under such a flow then

$$w(t) = \frac{K}{2} [1 + \tanh(\lambda_{\text{evo}} t + \omega_0)].$$

This is *hyperbolic kinematics*: constant speed in rapidity $\omega = \text{artanh}(2w/K - 1)$ and logistic dynamics in the native coordinate w .

We apply this structure to two domains. For evolution, the result yields the *Evolutionary Reality Equation* (ERE): linear motion in hyperbolic fitness rapidity and logistic trajectories in w . For measurement statistics, the same group law fixes the functional form of bounded, compositional observables; we obtain a *Hyperbolic Measurement Equation* (HME), an affine law in $\text{artanh} \delta$ versus an observer-capacity parameter Φ . This suggests—as a hypothesis for future work—a closed loop: evolutionary processes may select for efficient observers, whose measurement statistics reflect the same hyperbolic structure.

1 Introduction: Bounded Composition $\Rightarrow$ Hyperbolic Kinematics

Many real systems evolve within strict bounds (capacities, probabilities, performance ceilings). When successive changes compose associatively and the endpoints are invariant, admissible one-dimensional update laws are severely constrained. We formalize this and show that the unique linearizing coordinate is hyperbolic rapidity. The result recovers logistic/tanh dynamics in native coordinates and unifies evolutionary trajectories with measurement statistics.

Scope. This paper develops a *kinematics* for bounded one-dimensional composition. It does not claim universality beyond that scope. The Price equation is a coordinate-free decomposition; here we add structure: axioms that force a specific group law (Möbius/Einstein addition) and a unique linearizer (inverse tanh), yielding trajectories and falsifiable signatures.

2 Group-Theoretic Foundations and Uniqueness

2.1 Axioms

Let $u(t) \in (-1, 1)$ be a centered, scaled observable.¹ Assume:

(A1) Boundedness

fixed endpoints: the endpoints ± 1 are fixed points of the flow.

¹For $w \in (0, K)$ set $u = 2w/K - 1$.

- (A2) **Monotone C^1 flow:** $u(t)$ is C^1 and strictly monotone when the driving intensity is nonzero.
- (A3) **Associative, time-homogeneous composition:** updates over adjacent intervals compose via a C^1 associative binary law $\oplus$ on $(-1, 1)$.
- (A4) **Boundary-fixing homogeneity (Möbius):** each left translation $\tau_a(u) := a \oplus u$ extends to a real-analytic fractional linear map of the line that fixes ± 1 .

2.2 Uniqueness of the rapidity coordinate

Theorem 2.1 (Hyperbolic kinematics of bounded composition). *Under (A1)–(A4), there exists a strictly increasing bijection $f : (-1, 1) \rightarrow \mathbb{R}$ unique up to positive scaling such that*

$$u \oplus v = f^{-1}(f(u) + f(v)), \quad f(0) = 0. \quad (1)$$

Moreover, the boundary-fixing Möbius condition forces $f(u) = c \operatorname{artanh}(u)$ for some $c > 0$. Hence $u = \tanh(\omega)$ with rapidity $\omega := f(u)$, and any time-homogeneous flow is linear in ω .

Proof sketch. By Aczél’s representation theorem for associative C^1 laws on intervals, there exists an increasing f mapping $\oplus$ to $+$, unique up to scale. The Möbius hypothesis implies $\{\tau_a\}$ embeds into a one-parameter hyperbolic subgroup of $\operatorname{PSL}(2, \mathbb{R})$ fixing ± 1 , whose additive parameter is rapidity; the conjugating map is $f(u) = c \operatorname{artanh}(u)$. A complete derivation with explicit group-theoretic details is provided in a companion manuscript (in preparation).² $\square$

2.3 Physical interpretation and limitations of the axioms

Associativity (A3). Physically, associativity idealizes time-homogeneous composition: applying two selection intervals $\Delta t_1, \Delta t_2$ in either order yields the same net update in the coarse-grained variable. Real systems can deviate (context changes, frequency-dependent selection, non-Markov effects). In data, such deviations manifest as curvature or hysteresis in rapidity space; our tests below explicitly look for these failures.

Möbius homogeneity (A4). This encodes boundary-fixing symmetry and real-analyticity of translations. It is not independent dogma; it selects the hyperbolic subgroup compatible with fixed ± 1 . If the symmetry is broken (e.g., asymmetric headroom), alternative linearizers may arise; then the $\tanh$ signature should fail, which is itself diagnostic.

3 Hyperbolic Evolution Theory (ERE)

Let $w(t) \in (0, K)$ denote a bounded performance/fitness measure in a stationary environment over a window. Define

$$u := \frac{2w}{K} - 1 \in (-1, 1), \quad \omega := \operatorname{artanh}(u) \in \mathbb{R}, \quad (2)$$

so $u = \tanh \omega$ and $w = \frac{K}{2}(1 + \tanh \omega)$.

3.1 Constant drift in rapidity $\Leftrightarrow$ logistic in w

Assume piecewise-constant selection intensity: $d\omega/dt = \lambda_{\text{evo}}$ on each epoch. Differentiating $u = \tanh \omega$ gives

$$\frac{du}{dt} = \lambda_{\text{evo}} \operatorname{sech}^2 \omega = \lambda_{\text{evo}}(1 - u^2), \quad (3)$$

²Companion paper: *Einstein Velocity Addition from Associativity and Boundary Fixing*, in preparation.

thus

$$\frac{dw}{dt} = \frac{K}{2} \frac{du}{dt} = 2\lambda_{\text{evo}} w \left(1 - \frac{w}{K}\right), \quad (4)$$

which is the logistic ODE with rate $r = 2\lambda_{\text{evo}}$ and capacity K . Integrating,

$$\boxed{\omega(t) = \lambda_{\text{evo}}t + \omega_0 \iff w(t) = \frac{K}{2} \left[1 + \tanh(\lambda_{\text{evo}}t + \omega_0)\right]}. \quad (5)$$

3.2 Price equation mapping (mean form)

The continuous-time Price identity for trait z with Malthusian fitness m reads

$$\frac{d\bar{z}}{dt} = \text{Cov}(z, m) + \mathbb{E} \left[\frac{\partial z}{\partial t} \right]. \quad (6)$$

Choose $z = \omega(w)$ and assume (i) stationary environment (no explicit ∂_t), (ii) negligible transmission bias over the window, (iii) lineage-level dynamics follow (5). If all lineages share the same λ_{evo} within the epoch, then

$$\boxed{\frac{d\bar{\omega}}{dt} = \lambda_{\text{evo}}}. \quad (7)$$

Heterogeneity note: dispersion in lineage-specific λ_{evo} produces curvature in $\bar{\omega}(t)$; change-point and curvature tests can distinguish genuine violations of (5) from mixture effects.

3.3 Estimating K and robustness

In applications K is rarely known *a priori*. Treat K as a free parameter jointly fitted with $(\lambda_{\text{evo}}, \omega_0)$ via nonlinear regression, or estimate from independent ecological constraints. Modest misestimation of K perturbs $\omega = \text{artanh}(2w/K - 1)$ by an additive affine distortion; linearity tests in ω are typically robust provided the same K is used across the series.

3.4 Empirical signatures

- **Linearity in ω .** Plot $\omega(t) = \text{artanh}(2w/K - 1)$. Expect straight lines with slope changes only at environmental/constraint shifts.
- **Punctuated equilibrium.** Piecewise-constant λ_{evo} produces stitched tanh segments in $w(t)$.
- **Endpoint behavior.** Near $w \ll K$, $\tanh(\lambda_{\text{evo}}t + \omega_0) \approx \lambda_{\text{evo}}t + \omega_0$ (classical early-phase linear response).

3.5 Illustrative simulation (analytical curves)

4 Emergence of the Observer (minimal formalization)

Operational link. We posit that evolutionary architectures maximizing *progress per unit headroom* adopt dynamics close to constant rapidity drift. Define instantaneous fractional headroom $h(u) := 1 - u^2$ and note $du/dt = \lambda_{\text{evo}}h(u)$. Then $d\omega/dt = \lambda_{\text{evo}}$.

Proposition 4.1 (Efficiency of constant rapidity). *Among flows of the form $du/dt = \nu(t)h(u)$ with $|\nu(t)| \leq \nu_{\text{max}}$ and fixed endpoints, the time to reach any interior target in ω is minimized by constant $\nu(t) \equiv \nu_{\text{max}}$ (bang-bang optimality reduces to constant here). Thus selection for faster attainment of functioning thresholds favors approximately constant $d\omega/dt$.*

Sketch. The change of variables $\omega = \text{artanh}(u)$ yields $d\omega/dt = \nu(t)$. For a bound on $|\nu(t)|$, time-to-target in ω is minimized by taking $\nu(t)$ at its bound with constant sign; hence constant-speed geodesics in ω . $\square$

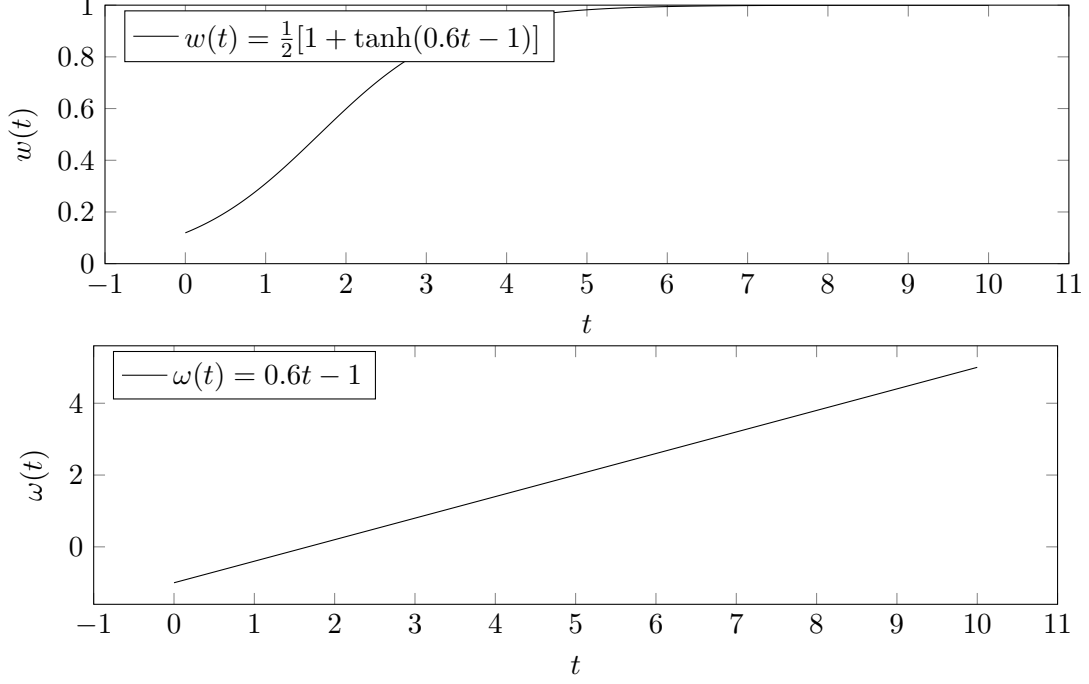


Figure 1: Analytical example: logistic trajectory in w corresponds to a straight line in rapidity ω .

Capacity parameter. Let Φ quantify effective information-throughput (e.g., bits/s relevant to the task, Fisher information rate, or a calibrated composite of physiological proxies). Empirically, higher Φ correlates with larger observed λ_{evo} in tasks where w is a performance proxy.

5 Quantum Measurement: Hyperbolic Measurement Equation (HME)

Let $\delta \in (-1, 1)$ be a bounded statistic (e.g., alternation-excess). Assume compositionality across independent contextual contributions and define rapidity $y := \text{artanh } \delta$. We posit

$$y(\Phi) = y_0 + \beta \Phi, \quad \beta > 0, \quad (8)$$

so that

$$\delta(\Phi) = \tanh(y_0 + \beta \Phi). \quad (9)$$

Why Φ is additive. In rapidity space, independent context contributions add. We interpret Φ as the cumulative, task-relevant information supplied to the measurement-basis setting; additivity is then a first-order approximation for concatenated preparation windows. (If additivity fails, (8) will be rejected by data.)

Parsimony. Writing (8) with a *single* slope β avoids redundant parameters ($2\alpha_0\kappa \mapsto \beta$). Identifiability of a product factorization can be addressed once domain-specific models separate α_0 and κ .

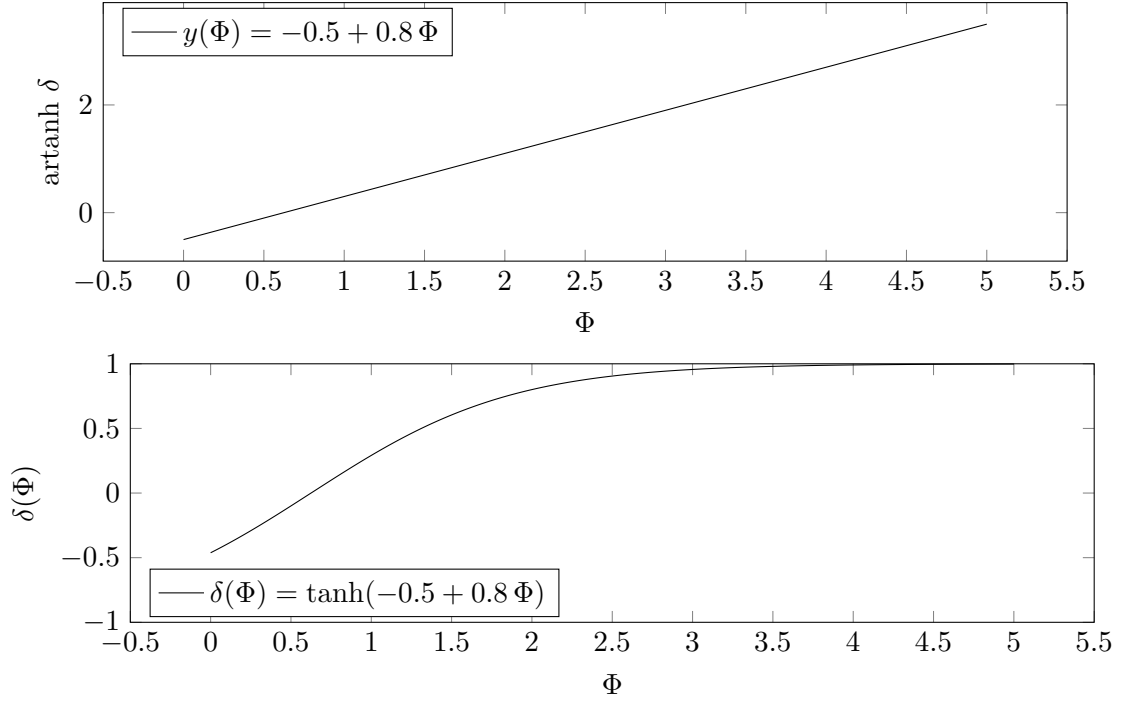


Figure 2: Hyperbolic Measurement Equation: affine rapidity (top) yields a tanh-shaped bounded statistic (bottom).

5.1 Illustrative analytic plot

6 Unified Loop and Predictions

6.1 Predictions and tests

P1 (ERE). In resource-bounded selection experiments with stable K , $w(t)$ fits (5); $\omega(t)$ is piecewise linear. Significant curvature in $\omega(t)$ beyond noise falsifies the ERE for that setting.

P2 (HME). In observer-QRNG (or analogous) protocols, $y = \text{artanh } \delta$ is affine in Φ with slope β . Model selection should decisively prefer the affine-in-rapidity model over alternatives at equal complexity.

6.2 Assumption checks and failure modes

- **Associativity test:** fit triads of intervals in permuted orders; detect non-associative residuals in ω .
- **Boundary symmetry:** asymmetry in approach speeds to 0 vs. K indicates broken Möbius homogeneity.
- **K uncertainty:** profile likelihood over K ; check stability of linear fits in ω .

7 Discussion

We have derived a kinematic law—hyperbolic linearity in rapidity—for bounded one-dimensional composition under explicit axioms, with applications to evolutionary trajectories (ERE) and measurement statistics (HME). The framework is falsifiable via linearity-in-rapidity tests. The proposed evolutionary–measurement *loop* is presented as a hypothesis for future work rather than a concluded causal mechanism. Key limitations include the idealization of associativity and boundary symmetry; we have outlined diagnostics for their violation.

Acknowledgments

We thank the mathematical literature on associative laws and Möbius actions for foundational results enabling Theorem 2.1.

A Sketch of Theorem 2.1

Let $\oplus$ be a C^1 associative law on $(-1, 1)$ with identity 0 and strictly increasing left translations. By Aczél, there exists increasing f such that $u \oplus v = f^{-1}(f(u) + f(v))$. The boundary-fixing homogeneity implies each τ_a is fractional linear fixing ± 1 , hence conjugate to a translation in $\mathbb{R}$ via f . The unique such conjugacy is $f(u) = c \operatorname{artanh}(u)$, $c > 0$. Therefore $u = \tanh(\omega)$ and time-homogeneous flows are linear in ω .

References

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